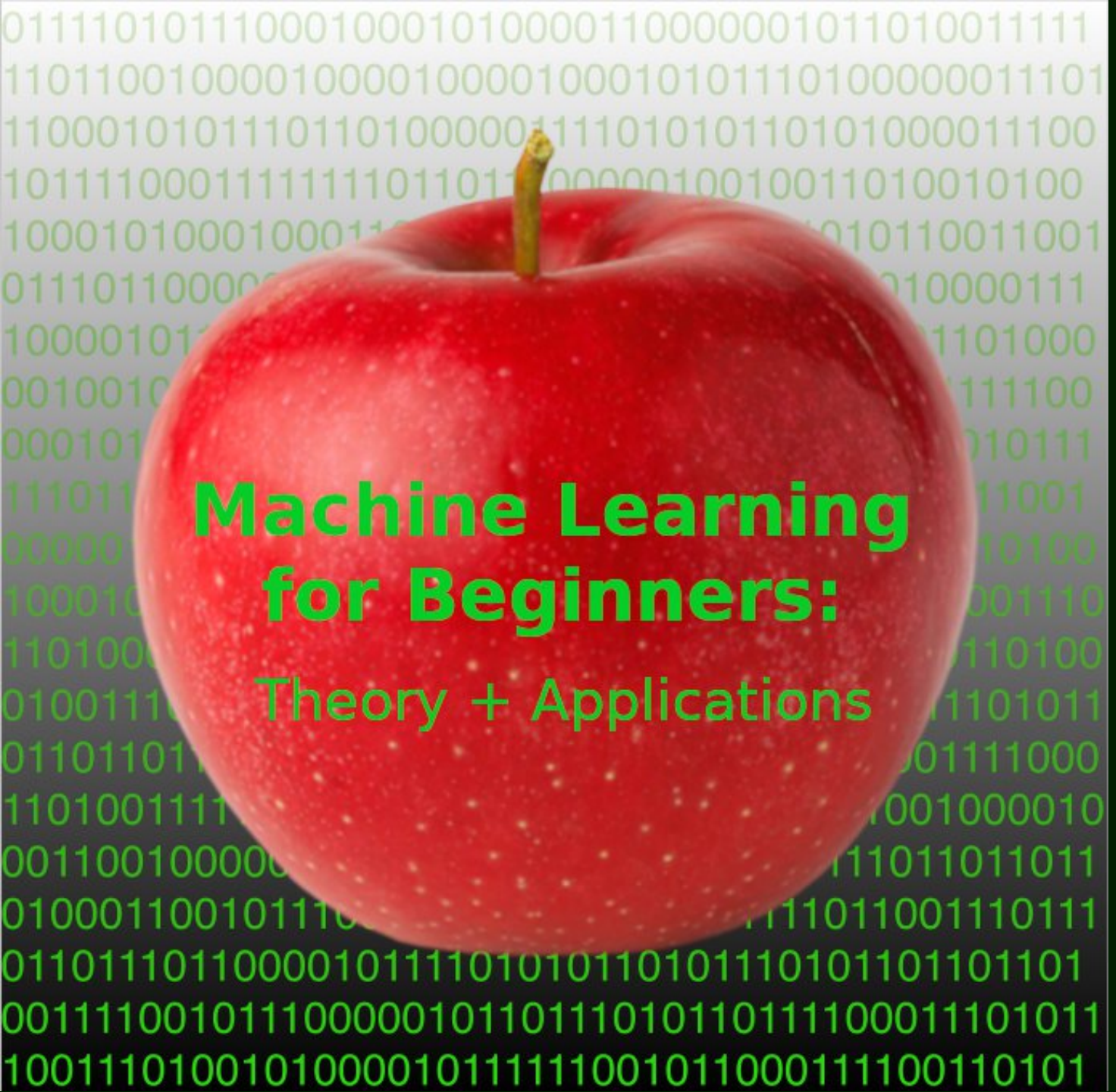



University of Bayreuth
Chair for Polymer Engineering
Prof. Dr. Holger Ruckdäschel

**Title:**

*"Machine Learning for Beginners: Theory and Applications"*

(Available in E-learning/Moodle)

**Lecturer:**

Dr. Rodrigo Q. Albuquerque

Lecture 5

# Topics



**Week 1: Introduction**
ML types, data, ethics, case studies

**Week 2: Preprocessing & *k*NN**
Preprocessing, *k*-nearest neighbors, distance metrics, hyperparameters, cross-validation

**Week 3: Linear Regression**
Types, loss function, (over/under)fitting bias/variance, L1/L2 regularization, metrics

**Week 4: Classification**
Logistic regression, SVM, classification evaluation metrics, confusion matrix one-hot/label encoding

★ **Week 5: Random forests**
Decision trees, random forests, classification and regression applications

**Week 6: PCA / KMeans**
Dimensionality reduction, clustering

**Week 7: Extra exercises**

**Week 8: Simple ANNs**
Feedforward networks, activation functions, basic training

**Week 9: Gaussian Processes**
Kernels, gaussian processes

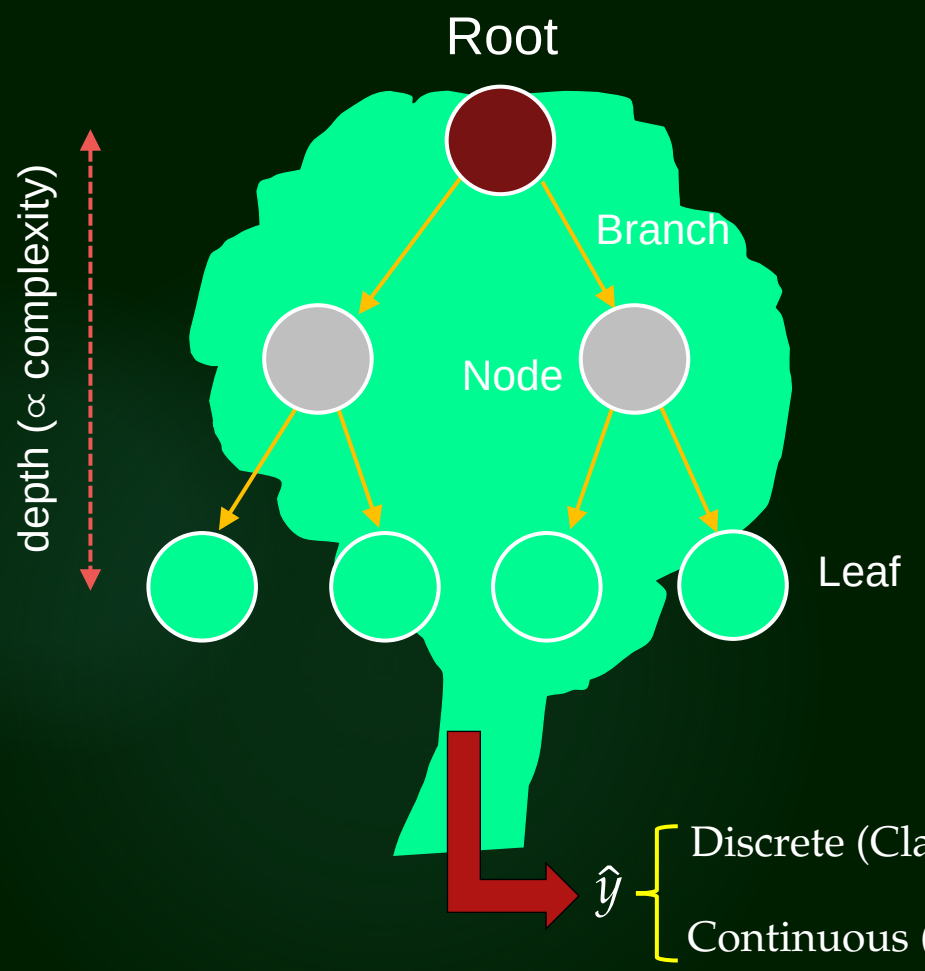
**Week 10: Design of Experiments**
DoE, random design, active learning, bayesian optimization

**Weeks 11-12: Student projects**

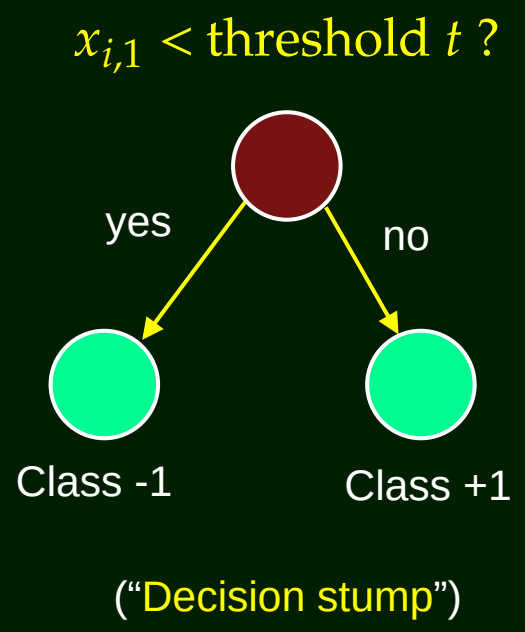
# Decision trees (DT)



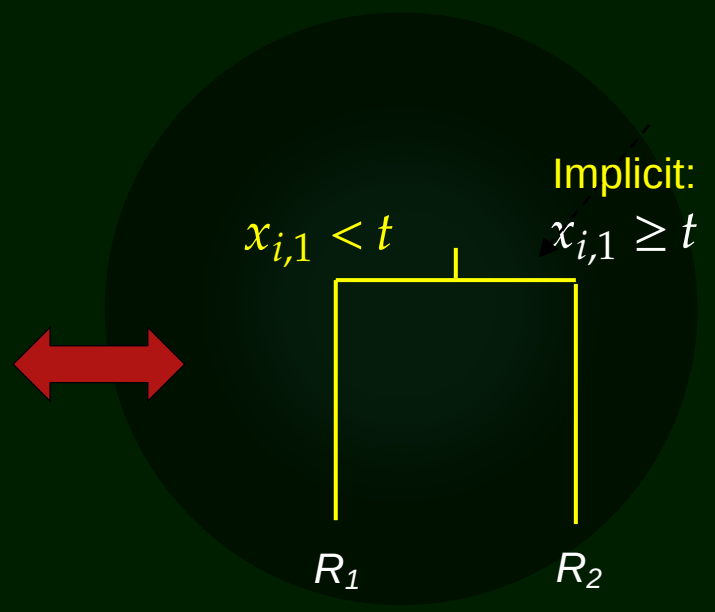
## Basics




$\hat{y}$ { Discrete (Classification) / Continuous (Regression) }




("Decision stump")

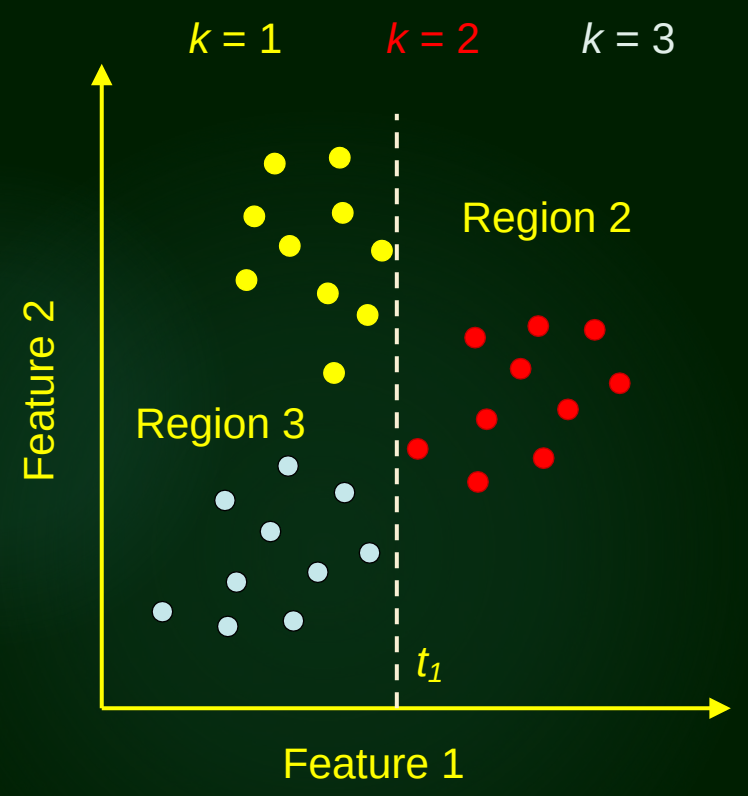



- How many nodes?
- Which features in which node?
- Which threshold to use?

# Decision trees (DT)



## Classification




Gini index (impurity measure) for region *m*:

$$G_m = \sum_{k=1}^{K} \hat{p}_{mk}(1-\hat{p}_{mk}) \qquad \hat{p}_{mk} = \frac{\#\ samples\ of\ class\ k\ in\ region\ m}{\#\ samples\ in\ region\ m}$$

Initial Gini index

whole data (region "1")

**=> first node (root)**

$$\begin{cases} \hat{p}_{11} = \hat{p}_{12} = \hat{p}_{13} = \frac{1}{3} = 0.33 \\ G_1 = 3\,[0.33\,(1-0.33)] = 0.67 \end{cases}$$

$t_1$ = one (of many!) possible thresholds for feature 1

$$\begin{cases} \text{Region 2:} \quad \hat{p}_{21} = \hat{p}_{23} = 0 \quad \hat{p}_{22} = 10/10 \quad G_2 = 1\,(1-1) = 0 \\ \text{Region 3:} \quad \hat{p}_{32} = 0 \quad \hat{p}_{31} = \hat{p}_{33} = 10/20 \quad G_3 = 2\,[0.5\,(1-0.5)] = 0.5 \end{cases}$$

Gini index variation ($\Delta G$)

$$\Delta G = G_{final} - G_{initial} = -10.1$$

Before splitting: $G_{initial} = G_1 \cdot n_1 = 0.67 \cdot 30\ samples$ = 20.1

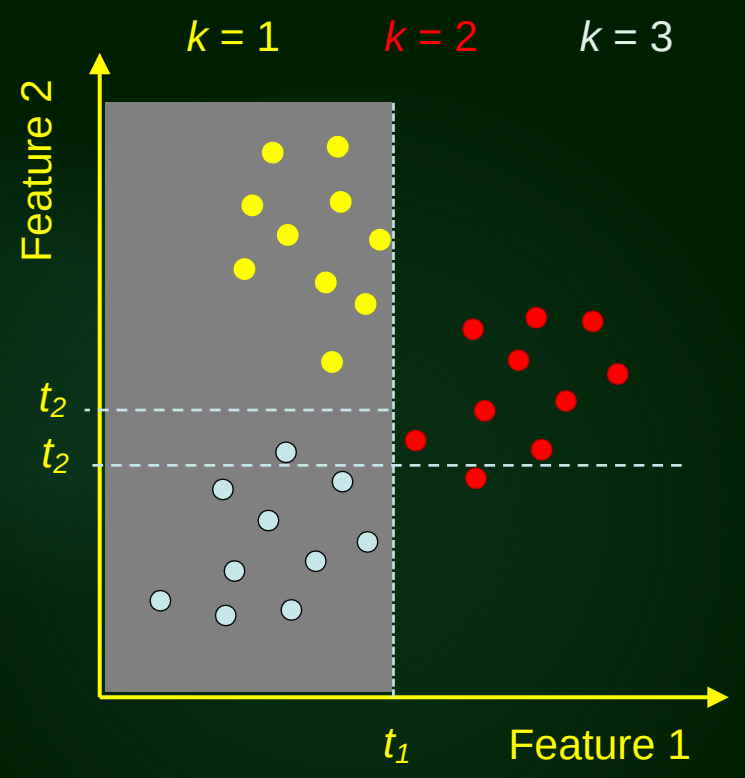
After splitting: $G_{final} = G_2 \cdot n_2 + G_3 \cdot n_3 = 0 \cdot 10\ samples + 0.5 \cdot 20 samples$ = 10

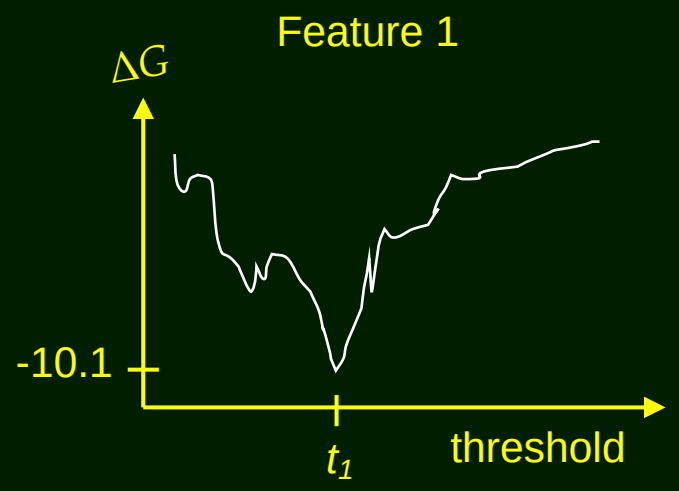
(Best threshold: Most negative $\Delta G$)

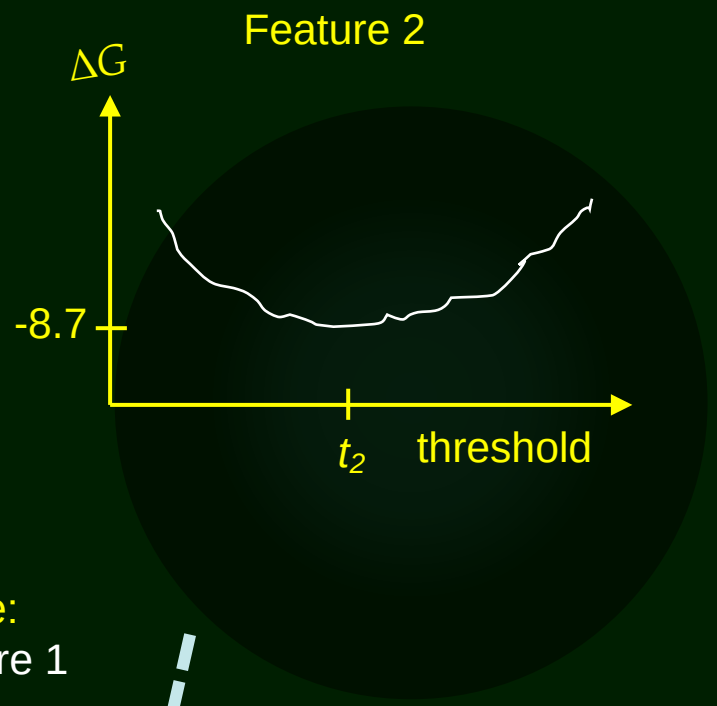
# Decision trees (DT)

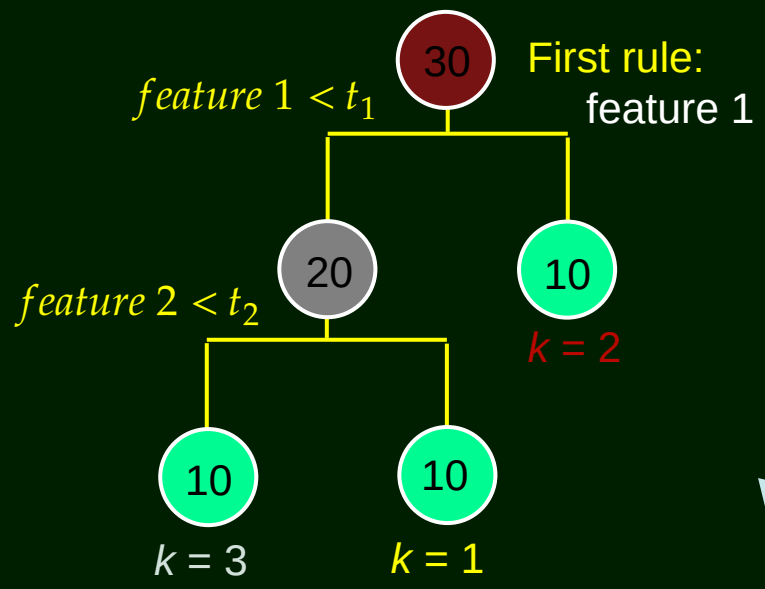


## Classification



$\hat{y}_m$ (leaf m):

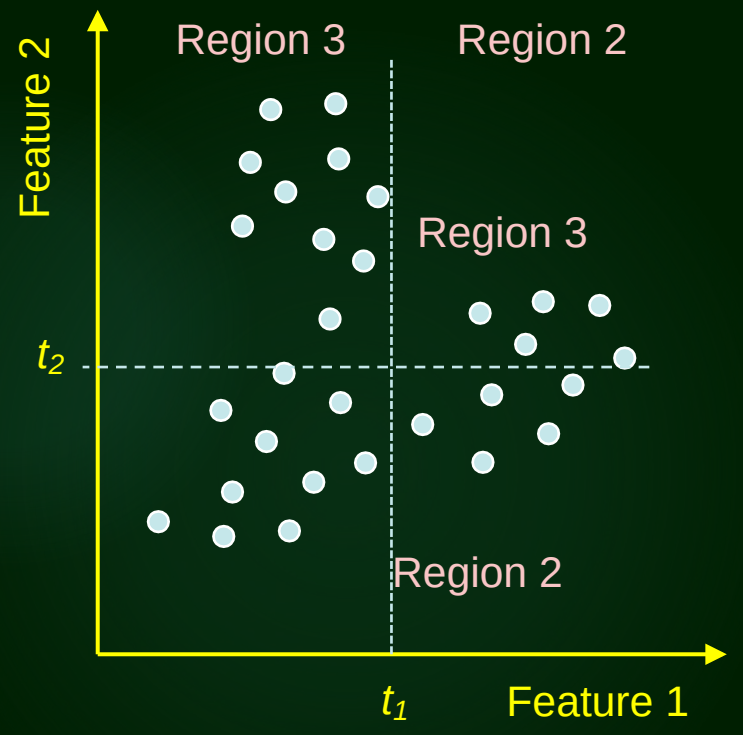
Majority class *k* *in node m*

# Decision trees (DT)

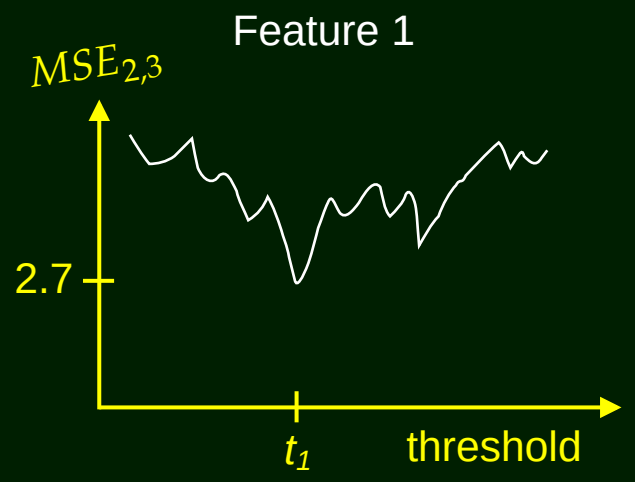


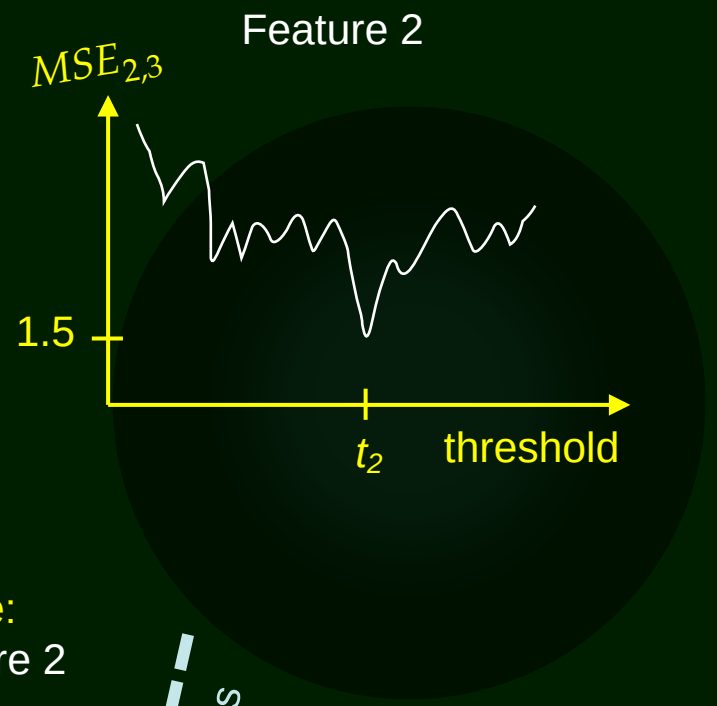
## Regression

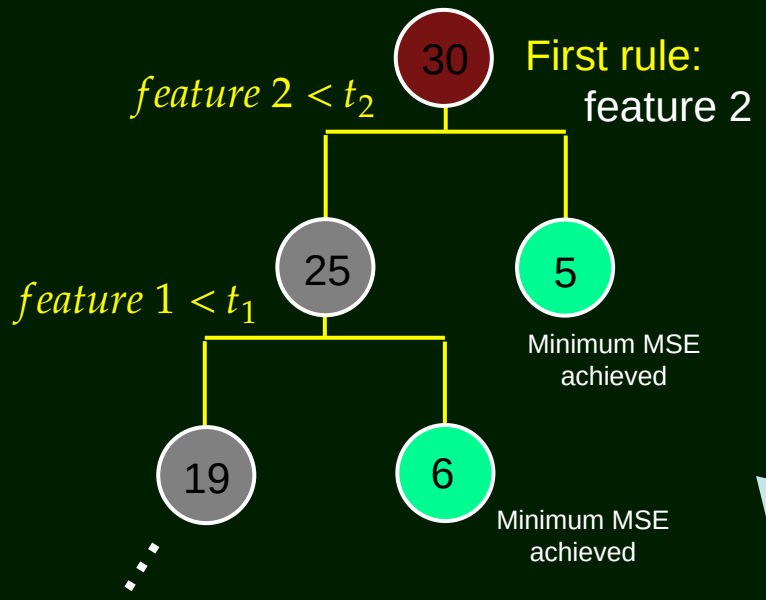
(Greedy approach)




$$MSE_{2,3} = MSE_2 + MSE_3$$




$MSE_{2,3}$ decreases

$\hat{y}_m$ (leaf m):

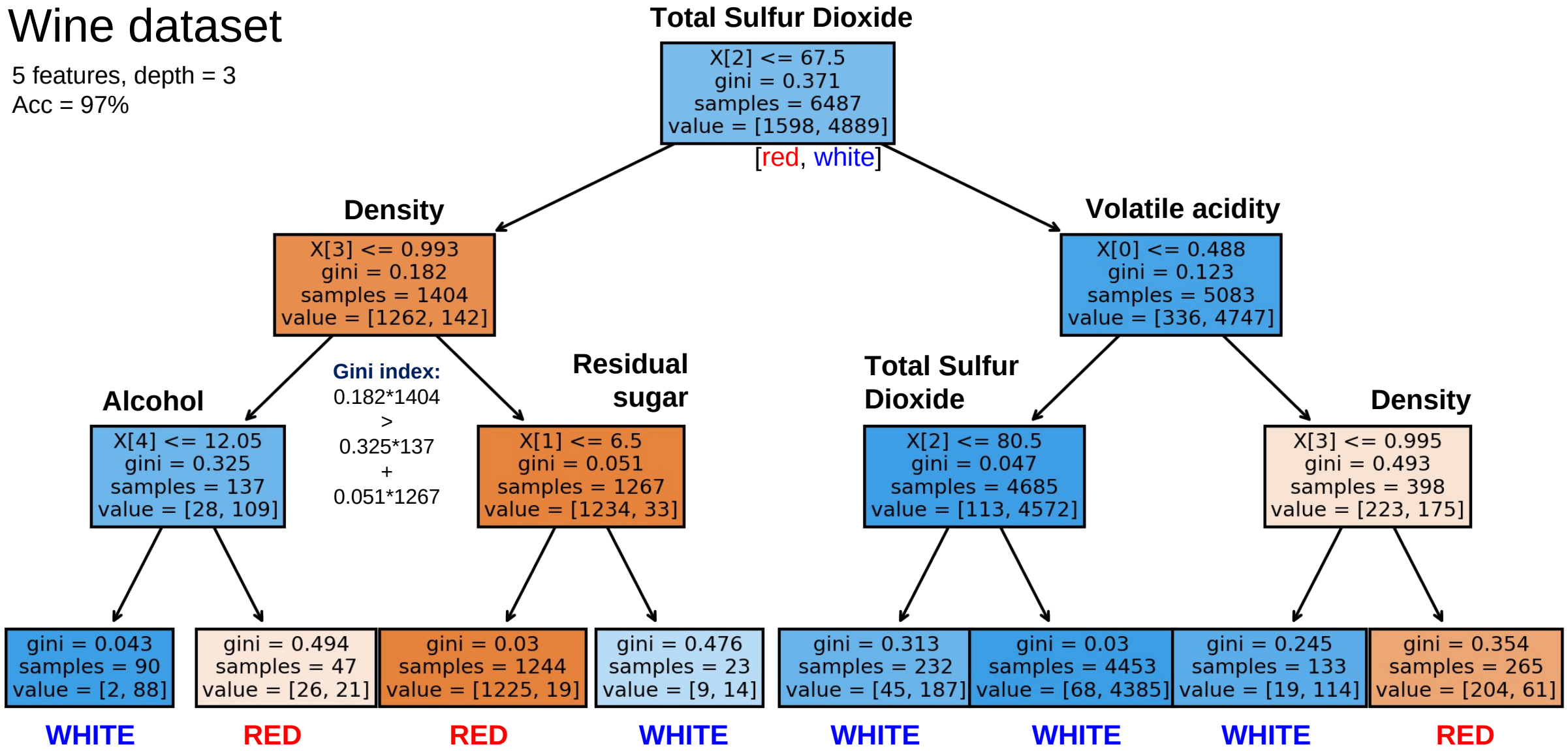
Average of target
*in node m*

# Decision trees (DT)



## Wine dataset

5 features, depth = 3
Acc = 97%



# Random Forests (RF)

**Problem:** Individual trees are still correlated!

**Solution:** *m* (random) features per split (*m* < *d*).



## Bootstrapping

$$\begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,d} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{n,d} \end{bmatrix}$$

*n* samples
*d* features

Training set *X*

Bootstrapping

Random selection of *n* samples
(with replacement!)

$B_1$ $B_2$ $B_3$ $\cdots$ $B_t$

*t* bootstrap sub-sets
(*n* samples and *d* features)

Build *1* model
(e.g. *1* tree)

$\hat{y}(X)$

Build *t* models
(e.g. *t* trees)

$\hat{y}(B_1)$ $\hat{y}(B_2)$ $\hat{y}(B_3)$ $\cdots$ $\hat{y}(B_t)$

Bagging
or "Bootstrap aggregation"

- Predictions are OK
- $\widehat{Var}[\hat{y}(X)] = ?$
  (no confidence interval)

$$\bar{y}^* = \frac{1}{t}\sum_{j=1}^{t} \hat{y}(B_j)$$

$$\widehat{Var}[\hat{y}(X)] = \frac{1}{t-1}\sum_{j=1}^{t} \left(\hat{y}(B_j) - \bar{y}^*\right)^2$$

# Random Forests (RF)

**NOTE:** RF is an ensemble method!



$$\begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,d} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n,1} & x_{n,2} & \cdots & x_{n,d} \end{bmatrix}$$

*n* samples
*d* features

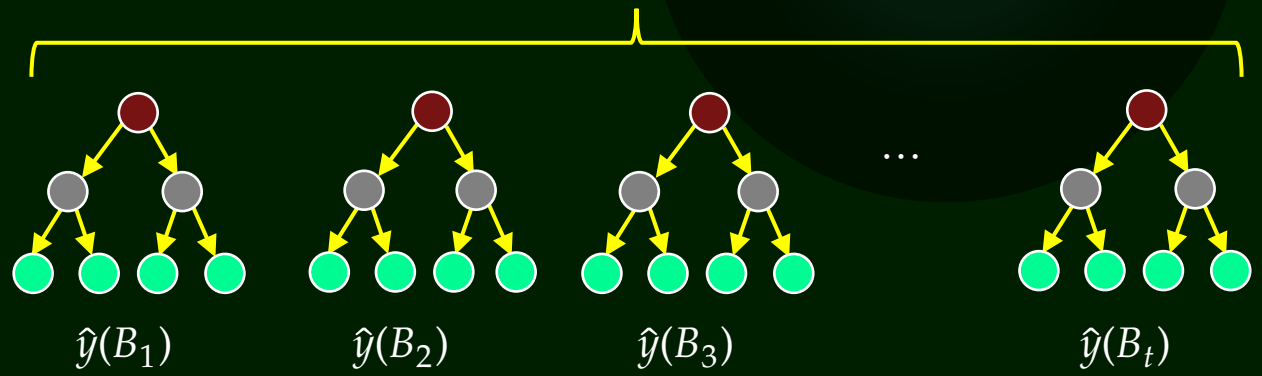
Training set *X*

Bootstrapping →

*t* (*uncorrelated*) trees

Each tree has:

- *m* randomly selected features (m < *d*)
- *n* randomly selected samples (with replacement)




Bagging
- Regression: average of individual regressions
- Classification: majority vote

**Out-Of-Bag (OOB) Error**

1) Make predictions for samples $x_i$ *(= OOB samples) using trees that don't contain them.*
2) Do this for all samples in *X*
3) Average errors (MSE, 1 - accuracy)

**Feature importances**

1) Record $\Delta G$ caused by each feature and in all trees
2) Weighted average (weight = # samples)
3) Normalize the averages, make a ranking

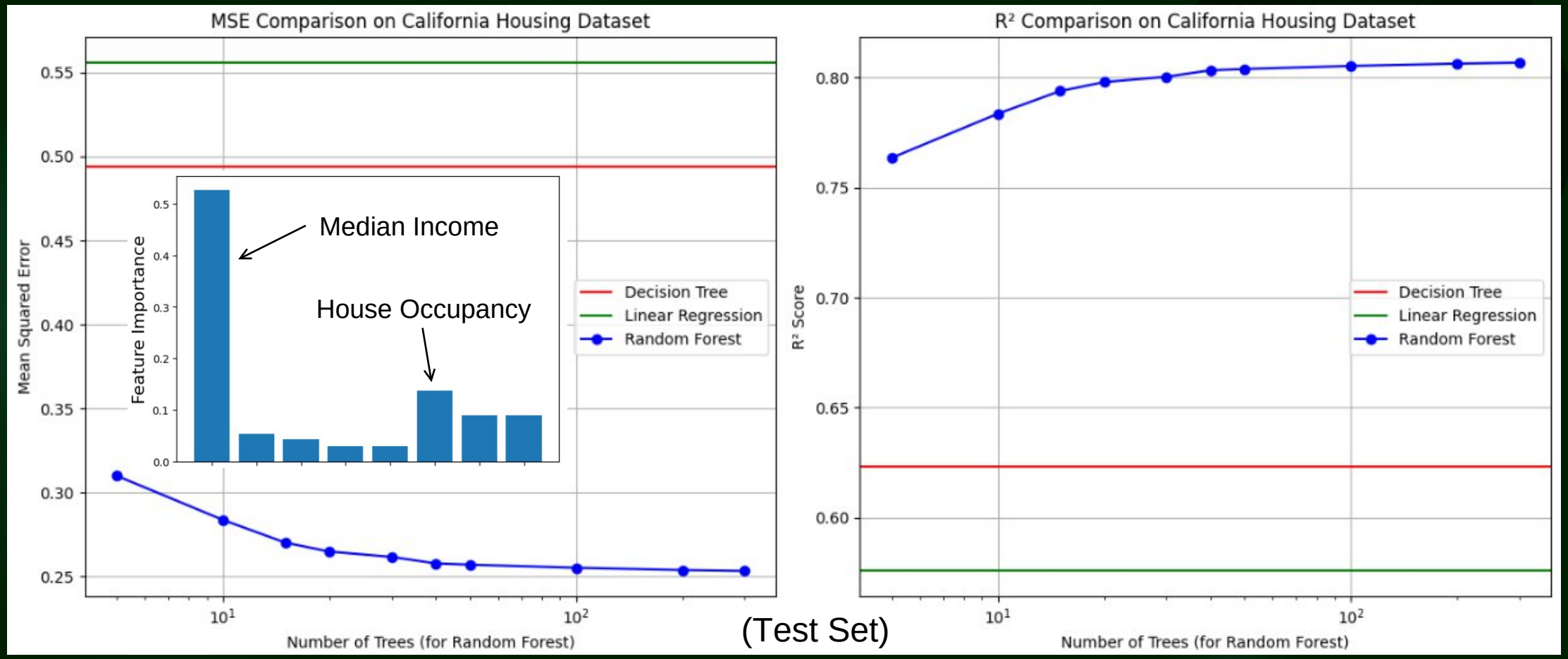
# Random Forests (RF)



## Predicting house prices

California house prices dataset (from sklearn), ca 20K houses

Features = ['Median Income', 'House Age', 'Ave Rooms', 'Ave Bedrms', 'Population', 'House Occup', 'Latitude', 'Longitude']




(Test Set)

# Other DT methods



### Gradient Boosting Machines (GBM)

Sequential ensemble learning method. Each tree corrects errors of preceding trees.
Trees built stagewise, optimizing a loss function. Popular implementations: XGBoost, LightGBM, CatBoost.

### HistGradientBoosting

Gradient boosting using histogram-based feature binning. Faster split finding during tree construction.
Can lead to faster training and lower memory usage.

### AdaBoost (Adaptive Boosting)

Weighted ensemble focusing on misclassified instances. Assigns higher weights to incorrectly predicted samples.
Effective for binary classification.

### Extremely Randomized Trees (Extra-Trees)

Similar to Random Forests with increased randomization. Uses random thresholds for feature splits.
Often faster to train than Random Forests.

### Isolation Forests

Specialized for unsupervised anomaly detection. Isolates outliers through random feature and split selection.
Anomalies require fewer splits to isolate.

# Exercises



1) Download the file ‘files5.zip’ from moodle/e-learning (folder “Week 5”) and unzip it locally on your computer
2) Go to google colab (online) **or** launch jupyter notebook locally (offline)
3) Open/upload ‘exercise5.ipynb’ and wait for more instructions